

Residual Diffusion Bridge Model for Image Restoration

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Abstract

Diffusion bridge models establish probabilistic paths between arbitrary paired distributions and exhibit great potential for universal image restoration. Most existing methods merely treat them as simple variants of stochastic interpolants, lacking a unified analytical perspective. Besides, they indiscriminately reconstruct images through global noise injection and removal, inevitably distorting undergraded regions due to imperfect reconstruction. To address these challenges, we propose the Residual Diffusion Bridge Model (RDBM). Specifically, we theoretically reformulate the stochastic differential equations of generalized diffusion bridge and derive the analytical formulas of its forward and reverse processes. Crucially, we leverage the residuals from given distributions to modulate the noise injection and removal, enabling adaptive restoration of degraded regions while preserving intact others. Additionally, we unravel the fundamental mathematical essence of existing bridge models, all of which are special cases of RDBM and empirically demonstrate the optimality of our proposed models. Extensive experiments are conducted to demonstrate the state-of-the-art performance of our method both qualitatively and quantitatively across diverse image restoration tasks. Code is publicly available at <https://github.com/MiliLab/RDBM>.

1. Introduction

Universal image restoration emerges as a unified paradigm integrating the perception, representation, and elimination of diverse degradations [19, 65], with the aim of restoring high-quality images from the degraded low-quality obser-

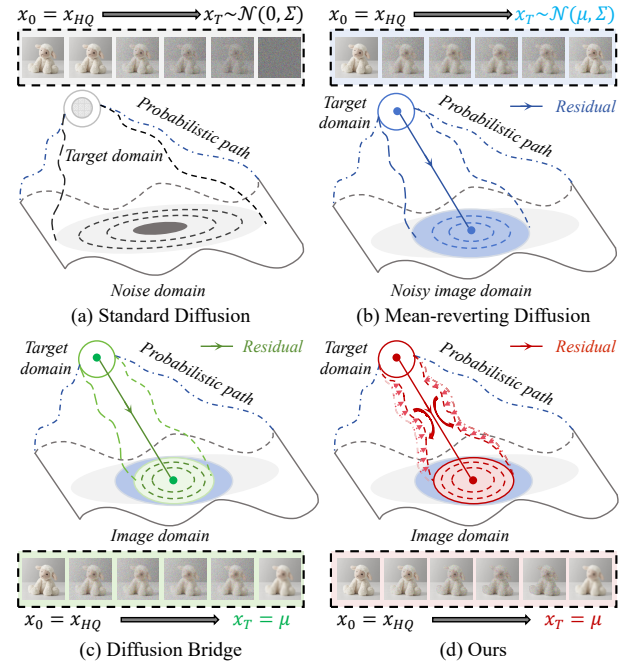


Figure 1. Typical diffusion processes. (a) Standard diffusion maps high-quality images to Gaussian noise domain. (b) Mean-reverting diffusion drives terminal state toward a low-quality domain with stationary noise. (c) Diffusion bridge establishes direct probabilistic transitions between known distributions. All inject noise globally, disrupting overall structures and constraining transitions. (d) In contrast, our RDBM selectively reconstructs degraded regions (e.g., the doll) while preserving intact areas (e.g., the background).

vation. It typically encompasses a broad range of classical tasks, including denoising, deraining, dehazing, super-resolving, and others [7, 14, 25, 64, 72]. Owing to the high fidelity of restored details, it has been widely adopted as a precursor across various downstream tasks [47, 65, 67, 83].

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Diffusion models have achieved remarkable advances in universal image restoration [43]. Early methods [62, 73, 89] follow the standard diffusion paradigm [16, 23] that maps images to a Gaussian distribution, and initializes the reverse inference from pure noise. Some approaches leverage generative priors pretrained from large models [40, 46] as conditional guidance for denoising networks. Others treat various restoration tasks as inverse problems by assuming access to degradation kernels [8, 9, 66, 79]. However, the randomness of noise and reliance on specific priors compromise both stability and universality. Subsequent studies [37, 41] incorporate mean-reverting dynamics into diffusion stochastic differential equations (SDEs), clustering forward terminal states around degraded observations to retain task-relevant cues. Additionally, diffusion bridges [85] directly model point-to-point stochastic transitions between paired distributions, thereby strengthening data associations and improving restoration fidelity. Despite these advances, existing methods still rely on global noise perturbation to construct probabilistic trajectories, requiring rigid reverse denoising processes, as shown in Fig. 1. However, they fail to distinguish regions with varying degradation levels and imperfectly reconstruct intact regions, limiting restoration performance and adaptivity. Besides, a systematic and theoretical framework is absent to elucidate the intricate interconnections among existing diffusion bridge formulations.

In this work, we propose a scalable and unified diffusion bridge framework for universal image restoration, termed Residual Diffusion Bridge Model (RDBM), and conduct a comprehensive analysis of optimal distribution transitions. Specifically, we integrate the mean-reverting property of the Ornstein–Uhlenbeck SDEs with Doob’s h -transform [58] to guide the terminal forward states toward degraded image distribution. Meanwhile, we use residuals from given distributions to dynamically modulate the probabilistic trajectories, thereby allowing the model to learn adaptive restoration of regions with varying degradation levels while mitigating redundant reconstruction in intact areas. Moreover, we theoretically demonstrate that our formulation yields the smooth distributional transition with respect to the residual-to-noise ratio. Building upon this, we uncover the mathematical essence of mainstream diffusion bridge formulations, all of which are special cases within our framework in specific configurations. Extensive experiments are conducted to verify the superiority of our method across diverse tasks including image restoration, translation, and inpainting. Our main contributions are summarized as follows:

1. We propose a scalable and unified diffusion bridge framework for image restoration. Theoretically, it is characterized as generalized stochastic interpolants that delineate probabilistic transitions between any paired distributions.
2. Benefiting from the certainty of terminal states, we exploit residuals from paired distributions to modulate noise

injection and removal, enabling selective reconstruction of degraded regions while preserving intact areas.

3. We unify and reinterpret existing bridge models as special instances of our RDBM framework, and substantiate its generality and effectiveness through extensive theoretical analysis and empirical validation.

2. Related Work

Denoising diffusion models [62, 63] were initially developed for image generation. Methods such as DiffIR [73], DvSR [71], and SR3 [77] directly repurpose diffusion models conditioned on degraded images for image restoration task, suffering from performance bottlenecks for task incompatibility. I2SB [35] and ColdDiffusion [5] bypass explicit noise perturbations and instead learn a degraded diffusion process directly via the network. Besides, RDDM [37], ResShift [76], ResFusion [59], and DiffUIR [84] incorporate prior distributions into the perturbation kernels to explicitly characterize degradation-aware diffusion processes. Moreover, IRSDE [41] employs a mean-reverting process to enforce diffusion trajectories that regress toward noisy degraded images with stationary variance. DDBM [85], BBDM [30] and GOUB [75] further apply Doob’s h -transform to remove terminal noise, offering a tractable alternative to pave the probability path that connects degraded and clean images, thereby achieving remarkable restoration performance. Flow matching [13, 33, 38] discards the stochastic noise and constructs the deterministic distribution transition path, thereby facilitating the optimal transport [88]. In contrast, our RDBM introduces the residual to modulate noise perturbation, enabling spatially adaptive restoration. Besides, RDBM can extend to these diffusion bridge models and flow matching in specific settings.

3. Background

3.1. Diffusion Bridge Models

Diffusion SDEs [16, 61] with drift term $\mathbf{f}(\cdot, t)$ and diffusion term $g(t)$ can be generally formulated as [24]:

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + g(t)d\omega_t, \quad (1)$$

where ω_t is the standard Wiener process. Eq. (1) describes the stochastic process from initial data $\mathbf{x}_0 \sim p_{data}(\mathbf{x})$ to a prior distribution $\mathbf{x}_T \sim p_{prior}(\mathbf{x})$. Its reverse SDEs and probability flow ordinary differential equations (ODEs) that share the same marginal distributions can be derived as:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)]dt + g(t)d\bar{\omega}_t, \quad (2)$$

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2}g^2(t)\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)]dt, \quad (3)$$

where $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ is score function. Furthermore, a diffusion process defined in Eq. (1) can be driven to arrive at a

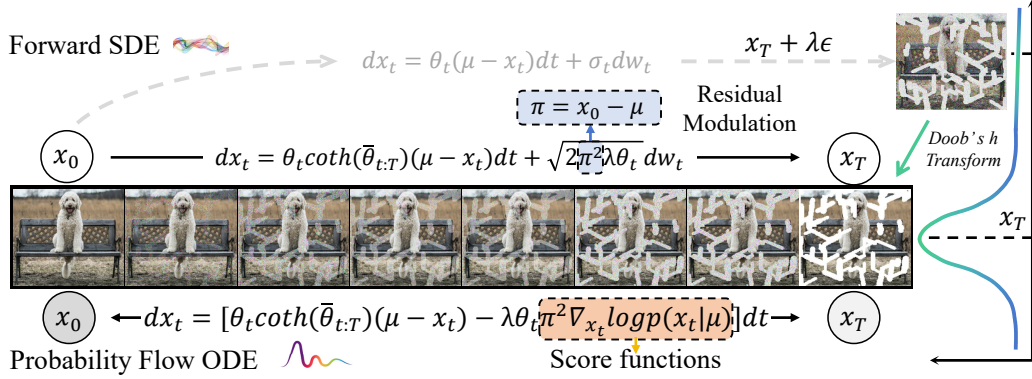


Figure 2. A schematic of Residual Diffusion Bridge Models. RDBM utilizes a diffusion process guided by Doob’s h -transform towards an endpoint $\mathbf{x}_T = \boldsymbol{\mu}$ free from stationary noise $\lambda\epsilon$. Modulated by the residual component $\boldsymbol{\pi} = \mathbf{x}_0 - \mathbf{x}_T$, the noise perturbation is selectively imposed on different regions with diverse degradation levels, thereby constructing probabilistic paths. Besides, it learns to reverse the process by matching the residual bridge score functions, facilitating an adaptive inversion from $\mathbf{x}_T \sim p_{\text{prior}}(\mathbf{x})$ to $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$.

particular point of interest $\boldsymbol{\mu}$ via Doob’s h -transform [56]:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) + g(t)^2 \mathbf{h}(\mathbf{x}_t, t, \mathbf{x}_T, T)]dt + g(t)d\omega_t, \quad (4)$$

where $\mathbf{h}(\mathbf{x}_t, t, \mathbf{x}_T, T) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_T | \mathbf{x}_t)$ is the gradient of the log transition kernel from t to T generated by the original SDE. When both the initial state $\mathbf{x}_0$ and terminal state $\mathbf{x}_T = \boldsymbol{\mu}$ are fixed, Eq. (4) defines a stochastic process known as a diffusion bridge (see proof in Suppl. A).

3.2. Ornstein Uhlenbeck Process

Ornstein–Uhlenbeck (OU) process is a stationary Gaussian–Markov process, with its marginal distribution converging toward a stable mean $\boldsymbol{\mu}$ with fixed variance over time. Formally, the OU process is generally defined as follows:

$$d\mathbf{x}_t = \theta_t(\boldsymbol{\mu} - \mathbf{x}_t)dt + \sigma_t d\omega_t, \quad (5)$$

where θ_t and σ_t respectively denote time-dependent drift and diffusion coefficients that characterize the speed of the mean-reversion. The transition probability of Eq. (5) admits a closed-form solution as below:

$$p(\mathbf{x}_t | \mathbf{x}_s) = \mathcal{N}(\bar{\mathbf{m}}_{s:t}, \bar{\sigma}_{s:t}^2 \mathbf{I}), \quad (6)$$

$$\mathcal{N}(\boldsymbol{\mu} + (\mathbf{x}_s - \boldsymbol{\mu})e^{-\bar{\theta}_{s:t}}, \int_s^t \sigma_z^2 e^{-2\bar{\theta}_{z:t}} dz) \quad (7)$$

$$\bar{\theta}_{s:t} = \int_s^t \theta_z dz. \quad (8)$$

Driven by the mean-reverting dynamics with Gaussian perturbations, the diffusion trajectory originates from $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ at initial time and gradually approaches $\mathbf{x}_T = \boldsymbol{\mu} \sim p_{\text{prior}}(\mathbf{x})$ at final time T . See Suppl. B for details.

4. Residual Diffusion Bridge Models

4.1. Generalized Forward Process

We redefine a OU process in Eq. (5) for generality:

$$d\mathbf{x}_t = \theta_t(\boldsymbol{\mu} - \mathbf{x}_t)dt + \pi \sigma_t d\omega_t, \quad (9)$$

where $\boldsymbol{\pi}$ is a predefined value. By applying the Doob’s h -transform to Eq. (9), we can establish the diffusion bridge that connects the high-quality image distribution $\mathbf{x}_0 \sim p_{\text{HQ}}(\mathbf{x})$ with degraded image distribution $\boldsymbol{\mu} \sim p_{\text{LQ}}(\mathbf{x})$:

Proposition 1 *Let $\mathbf{x}_t$ be a finite random variable governed by the generalized diffusion bridge process in Eq. (9), with terminal condition $\mathbf{x}_T = \boldsymbol{\mu}$. The evolution of its marginal distribution $p(\mathbf{x}_t | \mathbf{x}_T)$ satisfies the following SDE under a fixed drift-to-diffusion coefficient ratio $\lambda = \sigma_t^2 / (2\theta_t)$:*

$$d\mathbf{x}_t = \theta_t \coth(\bar{\theta}_{t:T})(\boldsymbol{\mu} - \mathbf{x}_t)dt + \sqrt{2\pi^2 \lambda \theta_t} d\omega_t, \quad (10)$$

where $\bar{\theta}_{s:t} = \int_s^t \theta_z dz$ and $\boldsymbol{\pi} \in \mathbb{R}$. See Suppl. C.

Consequently, Eq. (10) describes the generalized diffusion bridge models governed by λ , θ_t and $\boldsymbol{\pi}$. Here, λ controls the global noise level, while θ_t and $\boldsymbol{\pi}$ jointly determine bridge category and dynamical evolution. Furthermore, we can derive its closed-form solution as follows:

Proposition 2 *Given an initial state $\mathbf{x}_0$, the analytical solution of $\mathbf{x}_t$ at time $0 < t < T$ of that SDE in Eq. (10) can be formulated as:*

$$\mathbf{x}_t = \boldsymbol{\mu} + (\mathbf{x}_0 - \boldsymbol{\mu}) \frac{\sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{0:T})} + \int_0^t \sqrt{2\pi^2 \lambda \theta_s} \frac{\sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{s:T})} d\omega_s, \quad (11)$$

which satisfies a Gaussian distribution with expectation $E[\mathbf{x}_t]$ and variance $\text{Var}[\mathbf{x}_t]$ (proof is provided in Suppl. C):

$$E[\mathbf{x}_t] = \boldsymbol{\mu} + (\mathbf{x}_0 - \boldsymbol{\mu}) \frac{\sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{0:T})} := \boldsymbol{\mu} + (\mathbf{x}_0 - \boldsymbol{\mu}) \Theta_t, \quad (12)$$

$$\text{Var}[\mathbf{x}_t] = 2\pi^2 \lambda \frac{\sinh(\bar{\theta}_{0:t}) \sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{0:T})} := \pi^2 \Sigma_t^2. \quad (13)$$

Eq. (11) unveils that the trajectory of probability is dictated by a weighted amalgamation of the residual and Gaussian

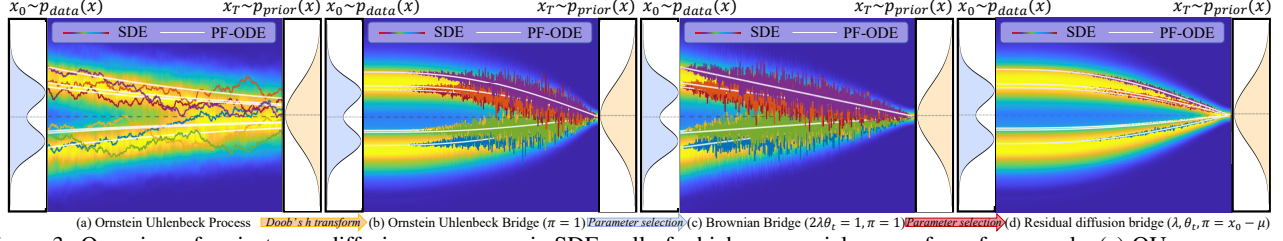


Figure 3. Overview of mainstream diffusion processes via SDEs, all of which are special cases of our framework. (a) OU process maps the data distribution to the prior distribution with noise. (b) OU bridge constructs probabilistic transition paths between given distributions. (c) Brownian bridge models linear expectations of intermediate states. (d) Our RDBM leverages residuals from paired distributions to adaptively modify the transition trajectories, maintaining a smooth residual-to-noise ratio.

noise. In order to delineate its temporal dynamic evolution, we define the residual-to-noise ratio (RNR) $R(t, i, j)$ for each pixel i, j at time t as follows (details are in Suppl. D):

$$R(i, j, t) = \frac{[x_0(i, j) - \mu(i, j)]^2}{2[\pi(i, j)]^2 \lambda} \frac{\sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{0:t}) \sinh(\bar{\theta}_{0:T})}, \quad (14)$$

which is governed by two terms. The first term depends on the residual component and fixed ratio λ . The second term is entirely determined by θ_t series and exhibits a monotonic decline, which diverges to infinity as time $t \rightarrow 0$ and converges to an infinitesimal value as time $t \rightarrow T$. Previous works [41, 75, 85] typically set $\pi = 1$, thereby performing the global noise perturbation to uniformly disrupt the overall structure of images. This induces two ill-posed issues: (i) degraded regions with varying levels are treated equally, and intact regions suffer redundant and imperfect reconstruction due to inevitable cumulative error in reverse process. (ii) pixel-wise numerator $[x_0(i, j) - \mu(i, j)]^2$ may exhibit discontinuous jumps, potentially distorting the smooth monotonic decay of the residual-to-noise ratio. Therefore, to maintain the dynamic equilibrium in transmission trajectories, we fix $\pi = \mathbf{x}_0 - \mu$, thereby deriving our specific formulation within this framework with adaptive noise perturbation and pixel-independent residual-to-noise ratio:

$$R(t, i, j) = R(t) \propto \frac{\sinh(\bar{\theta}_{t:T})}{\sinh(\bar{\theta}_{0:t}) \sinh(\bar{\theta}_{0:T})}. \quad (15)$$

4.2. Reverse Process and Training Objective

From Eq. (12)-(13), the transition probability distributions from initial state $\mathbf{x}_0$ to intermediate states $\mathbf{x}_t$ and $\mathbf{x}_{t-1}$ are:

$$q(\mathbf{x}_t | \mathbf{x}_0, \mu) = \mathcal{N}(\mu + (\mathbf{x}_0 - \mu)\Theta_t, \pi^2 \Sigma_t^2 \mathbf{I}), \quad (16)$$

$$q(\mathbf{x}_{t-1} | \mathbf{x}_0, \mu) = \mathcal{N}(\mu + (\mathbf{x}_0 - \mu)\Theta_{t-1}, \pi^2 \Sigma_{t-1}^2 \mathbf{I}), \quad (17)$$

Supposing that sampling from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ follows the Gaussian distribution, we leverage *Bayes' theorem* to derive the

Table 1. Connections to other mainstream bridge models.

Diffusion Bridge Configurations			Method
$\theta_t \rightarrow 0$	λ	$\pi = 0$	Flow Matching [33, 38]
$\theta_t \rightarrow 0$	$\lambda \rightarrow \infty$	$\pi = 1$	VE Bridge [85]
$\theta_t \rightarrow 0$	$\lambda \rightarrow \frac{1}{2}$	$\pi = 1$	VP Bridge [85]
$\theta_t \rightarrow 0$	$2\lambda\theta_t \rightarrow 1$	$\pi = 1$	Brownian Bridge [30, 35]
θ_t	λ	$\pi = 1$	OU Bridge [75]
θ_t	λ	$\pi = \mathbf{x}_0 - \mu$	Ours

deterministic sampling of reverse process (see Suppl. E):

$$\mathbf{x}_{t-1} = \mu + \frac{\Sigma_{t-1}}{\Sigma_t}(\mathbf{x}_t - \mu) + (\Theta_{t-1} - \Theta_t \frac{\Sigma_{t-1}}{\Sigma_t})\pi, \quad (18)$$

$$= \mu + \frac{\Theta_{t-1}}{\Theta_t}(\mathbf{x}_t - \mu) - (\frac{\Theta_{t-1}}{\Theta_t} \Sigma_t - \Sigma_{t-1})\pi \epsilon_t, \quad (19)$$

Apparently, Eq. (19) involves two unknowns, the residual π and the noise ϵ_t . In theory, the distributions at all time steps should be aligned; thus, the overall training objective is:

$$\mathcal{L}(\dot{\theta}) = D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0, \mu) || p_{\dot{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mu)). \quad (20)$$

Assuming $p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mu)$ follows a Gaussian distribution centered at $m_{\dot{\theta}}(\mathbf{x}_t, \mathbf{x}_0)$ with a constant variance, minimizing the Kullback-Leibler divergence [26] D_{KL} is equivalent to reducing the distance between the means (see Suppl. G):

$$\mathcal{L}(\dot{\theta}) := \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)}[\eta_m \|m(\mathbf{x}_t, \mathbf{x}_0) - m_{\dot{\theta}}(\mathbf{x}_t, \mathbf{x}_0)\|] \quad (21)$$

$$:= \mathbb{E}_{\mathbf{x}_0, \mu, t}[\eta_{\epsilon} \|\pi_{\epsilon}^{\dot{\theta}}(\mathbf{x}_t, t, \mu) - (\mathbf{x}_0 - \mu)\epsilon_t\|], \quad (22)$$

where η_m and η_{ϵ} are different weights for different training objectives. Accordingly, we can employ a neural network $\pi_{\epsilon}^{\dot{\theta}}(\mathbf{x}_t, t, \mu)$ to predict the multiplication of residual and noise at once. The detailed algorithms for training and sampling are presented in Alg. 1 and Alg. 2, respectively.

4.3. Analysis

We redefine a general mean-reverting process in Eq. (9) and employ Doob's h transform to derive the generalized diffusion bridge in Eq. (10) that exhibits the property of

Algorithm 1: Training.

Input: Clean image $\mathbf{x}_0$; Degraded image: μ ;
Residual map: $\pi = \mathbf{x}_0 - \mu$.

```

1 repeat
2    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ ;
3    $t \sim \text{Uniform}(1, \dots, T)$ ;
4    $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ ;
5    $\mathbf{x}_t = \mu + (\mathbf{x}_0 - \mu)\Theta_t + \pi\Sigma_t\epsilon$ ;
6   Take the gradient descent step on
7      $\nabla_{\theta} \|\pi\epsilon - \pi_{\epsilon}^{\theta}(\mathbf{x}_t, t, \mu)\|_1$ 
8 until converged;
```

mean-arrival. Probability paths of several diffusion processes are shown in Fig. 3. We configure π to serve as residual component for adaptive noise perturbation, yielding a smoothly decaying RNR. Besides, other mainstream bridge models can be concluded in our framework, such as standard diffusion bridge [85], Brownian Bridge [30, 35], OU Bridge [75], Flow Matching [33, 38] and others, as summarized in Tab. 1 (See Suppl. F).

5. Experiments

5.1. Datasets and Evaluation Metrics

Extensive experiments are conducted to assess the performance of our method on five image restoration tasks, including deraining, low-light enhancement, desnowing, dehazing, and deblurring. For fairness, we collect and mix the most widely used datasets for each task as follows. Besides, dataset details are as summarized in Suppl. H.

Image deraining. We train our model on the merged datasets from Rain13K [20] and DeRaindrop [52], which cover diverse rain streaks and densities. Evaluation is conducted on both rain- and raindrop-removal tasks using the mixed datasets [32, 52, 74, 80]. In addition, we assess zero-shot generalization on real-world datasets, including GT-Rain [4] without ground-truth for reference.

Low-light enhancement. We combine the LOL [70] and VE-LOL-L [36] datasets, which furnish real and synthetic paired samples across diverse scenes with varying illumination and noise levels. Additionally, we employ the NPE [68], MEF [45] and DICM [27] datasets to conduct zero-shot generalization on real-world scenarios.

Image desnowing. We adopt the CSD [6] dataset as the primary benchmark and evaluate real-world performance on Snow100K-Real [39], which has no ground-truth.

Image dehazing. We adopt ITS_v2 [28] and D-HAZY [10] as training benchmarks, encompassing diverse scenes under varying haze densities. The outdoor subset SOTS [28] is used for evaluation, while real-world generalization is assessed on Dense-Haze [2], NHRW [81], and NH-HAZE [3].

Algorithm 2: Sampling.

Input: Degraded image: μ ; Neural network $\pi_{\epsilon}^{\theta}(\cdot)$.

```

1 for  $t = T$  to 1 do
2    $\pi\epsilon = \pi_{\epsilon}^{\theta}(\mathbf{x}_t, t, \mu)$ 
3   if  $t = T$  then
4      $\mathbf{x}_{T-1} = \mu$ 
5   else
6      $\mathbf{x}_{t-1} = \mu + \frac{\Theta_{t-1}}{\Theta_t}(\mathbf{x}_t - \mu) - (\frac{\Theta_{t-1}}{\Theta_t}\Sigma_t - \Sigma_{t-1})\pi\epsilon$ 
7 end
Output:  $\mathbf{x}_0$ .
```

Image deblurring. We use the GoPro [49] dataset to perform deblurring tasks, which contains various levels of blur obtained by averaging the clear images captured in very short intervals. To further validate the generalizability, we perform zero-shot restoration on the RealBlur [54] dataset.

Benchmarks are evaluated using peak signal-to-noise ratio (PSNR) [17], structural similarity (SSIM) [69], natural image quality evaluator (NIQE) [27] in RGB space, and the learned perceptual image patch similarity (LPIPS) [60] in feature space. For fairness, we compare our method with several universal restoration methods [11, 12, 21, 29, 31, 41, 42, 44, 51, 53, 75, 78, 82], which are all re-implemented on the mixed datasets for comparisons.

5.2. Implementation Details

Our method is trained using 8 Nvidia A800 GPUs with PyTorch [50] framework for 128h. Adam optimizer and L1 loss are employed for 500k iterations with a learning rate of $1e-4$. We set the batch size as 20 and distribute it evenly to each task. We randomly crop patches of size 256×256 from the original image as network input for training and use 10 timesteps for full-resolution testing. We utilize U-Net [57] architecture as network backbone. We change the channel number of the hidden layers C to obtain different versions with varied parameter quantities:

- RDBM-T: $C=32$, channel multiplier = $\{1, 1, 1, 1\}$
- RDBM-S: $C=32$, channel multiplier = $\{1, 2, 2, 4\}$
- RDBM-B: $C=64$, channel multiplier = $\{1, 2, 2, 4\}$
- RDBM-L: $C=64$, channel multiplier = $\{1, 2, 4, 8\}$

5.3. Comparative Experiments

We compare our RDBM with several representative universal methods across five challenge image restoration tasks.

Visual comparison. The qualitative results are illustrated in Fig. 4. For more results, please refer to Suppl. H. Obviously, our method generates high-quality results that are the most similar to ground-truth compared with other methods.

Quantitative evaluation. We present quantitative results in Tab. 2. Clearly, RDBM-L attains great performance im-

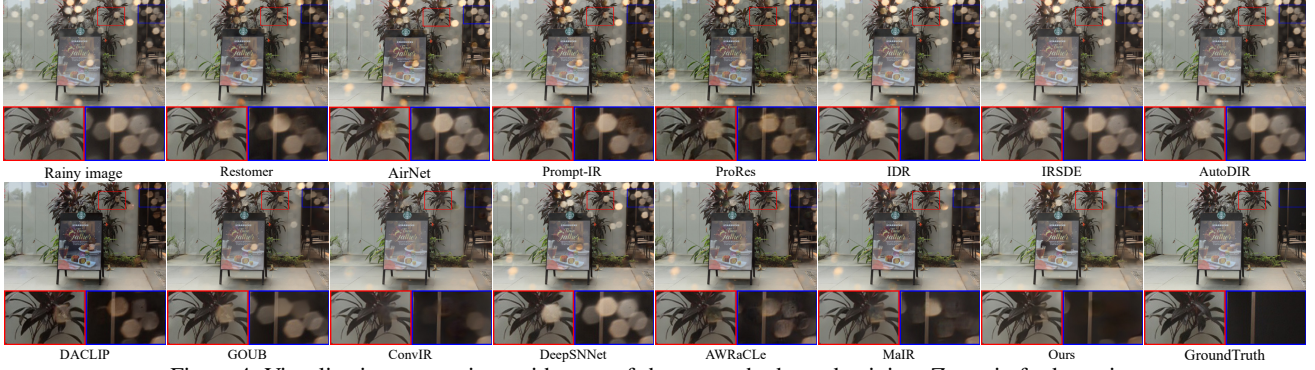


Figure 4. Visualization comparison with state-of-the-art methods on deraining. Zoom in for best view.

Table 2. Quantitative comparisons of five image restoration tasks. The FLOPS is calculated in the inference stage with 256×256 resolution. The best and second best results of universal models are shown in **red** and **blue**, respectively.

Method	Year	Deraining		Enhancement		Desnowing		Dehazing		Deblurring		Average		Complexity	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	Params(M)	FLOPs(G)
Restormer [78]	2022	28.54	0.847	21.75	0.742	28.53	0.919	26.54	0.924	26.44	0.799	27.61	0.869	26.09	140.99
AirNet [29]	2022	24.78	0.774	13.05	0.485	25.80	0.885	18.53	0.827	25.76	0.782	24.01	0.809	5.76	301.27
Prompt-IR [51]	2023	28.97	0.856	20.97	0.733	29.52	0.938	25.80	0.929	26.25	0.797	27.89	0.878	32.96	158.14
ProRes [44]	2023	22.42	0.752	20.31	0.741	24.53	0.859	24.81	0.888	26.08	0.792	24.08	0.814	370.26	97.17
IDR [82]	2023	28.40	0.844	20.95	0.706	27.77	0.911	24.48	0.914	26.33	0.799	26.96	0.863	6.19	32.16
IRSDE [41]	2023	24.05	0.822	11.29	0.450	15.91	0.806	11.52	0.697	26.68	0.811	19.55	0.783	137.13	379.33
AutoDIR [21]	2024	29.32	0.863	15.65	0.707	15.31	0.706	19.01	0.829	28.47	0.864	22.43	0.799	115.86	63.38
DA-CLIP [42]	2024	28.63	0.854	19.50	0.730	28.23	0.934	27.26	0.941	26.47	0.818	27.54	0.881	32.96	158.14
GOUB [75]	2024	28.65	0.870	17.80	0.723	30.39	0.960	20.85	0.902	27.85	0.838	27.60	0.895	137.13	379.34
ConvIR [11]	2024	29.18	0.867	21.36	0.771	31.43	0.950	29.13	0.960	28.41	0.862	29.49	0.903	14.82	128.93
DeepSNNNet [12]	2025	28.62	0.845	17.90	0.661	30.02	0.927	28.72	0.937	25.81	0.773	28.15	0.865	17.32	71.79
AWRaCLe [53]	2025	29.15	0.860	20.41	0.756	27.70	0.927	18.38	0.789	26.37	0.818	26.31	0.861	94.18	165.42
MaIR [31]	2025	29.45	0.864	21.76	0.750	30.80	0.955	30.39	0.960	28.28	0.859	29.51	0.904	20.71	110.44
RDBM-T	-	27.98	0.844	21.04	0.745	28.47	0.918	26.88	0.928	25.82	0.784	27.31	0.865	0.45	5.74
RDBM-S	-	29.23	0.864	21.98	0.765	30.93	0.941	28.92	0.942	26.67	0.808	28.99	0.886	1.07	8.01
RDBM-B	-	29.70	0.875	22.00	0.761	32.48	0.956	31.56	0.966	27.81	0.842	30.24	0.904	3.65	23.97
RDBM-L	-	30.31	0.884	24.53	0.812	32.59	0.961	33.45	0.965	29.04	0.877	31.04	0.917	7.73	32.93

Table 3. Performance of different noise schedule ($\lambda = 10/255$).

Schedule	Deraining		Enlightening		Desnowing		Dehazing		Deblur		Average	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Linear	29.63	0.878	22.39	0.774	34.15	0.965	32.00	0.958	28.48	0.864	30.99	0.912
Cosine	30.31	0.884	24.53	0.812	32.59	0.961	33.45	0.965	29.04	0.877	31.04	0.917
Sigmoid	29.31	0.868	22.91	0.782	33.80	0.961	32.06	0.970	28.63	0.869	30.84	0.911

Table 4. Performance of varied stationary variance λ .

λ	Deraining		Enlightening		Desnowing		Dehazing		Deblur		Average	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
1/255	30.03	0.884	23.87	0.823	31.86	0.957	31.27	0.959	28.89	0.874	30.36	0.915
10/255	30.31	0.884	24.53	0.812	32.59	0.961	33.45	0.965	29.04	0.877	31.04	0.917
20/255	29.86	0.879	22.20	0.756	33.40	0.962	31.52	0.966	28.08	0.850	30.66	0.909
50/255	29.94	0.884	22.61	0.782	32.66	0.964	29.51	0.950	28.55	0.867	30.26	0.913
100/255	29.98	0.880	23.49	0.794	30.09	0.950	27.03	0.940	28.53	0.865	29.08	0.906

provement across all tasks by a large margin, culminating in average gains of 1.55 dB in PSNR and 0.013 in SSIM. For fair comparisons, we also evaluate several lightweight RDBM variants. Notably, RDBM-B also attains the best average metrics with fewer parameters, highlighting the effectiveness of our design. Moreover, our models exhibit high scalability across different parameter levels. In conclusion, our method outperforms others and is the most competitive.

5.4. Ablation Study

To thoroughly explore the efficacy of our method, we carry out ablation studies encompassing three distinct categories: (i) the influence of various implementation configurations, (ii) the performance across different sampling steps, and (iii) the impact of diverse diffusion bridge settings.

Influence of various implementation configurations. Our RDBM formulations are governed by the schedule $\{\theta_t\}$

and stationary variance λ . Initially, we adopt the empirical choice $\lambda = \frac{10}{255}$ [75] and compare performance across different noise schedules, as reported in Tab. 3. It is evident that the optimal noise schedules differ for distinct restoration tasks, with the cosine schedule generally yielding the best results. Building on this finding, we further conduct a quantitative comparison among diverse stationary variance λ , as presented in Tab. 4. The results indicate that $\lambda = \frac{10}{255}$ with cosine noise schedule is the optimal configuration.

Performance across different sampling steps. Model efficiency and restoration quality hinge on the sampling steps, quantified by neural function evaluations (NFEs). We provide the restoration performance of different sampling steps in Tab. 5. Clearly, our model exhibits varying performance across different NFEs. Initially, the restoration performance increases with more steps and peaks at 10 NFEs, reflect-

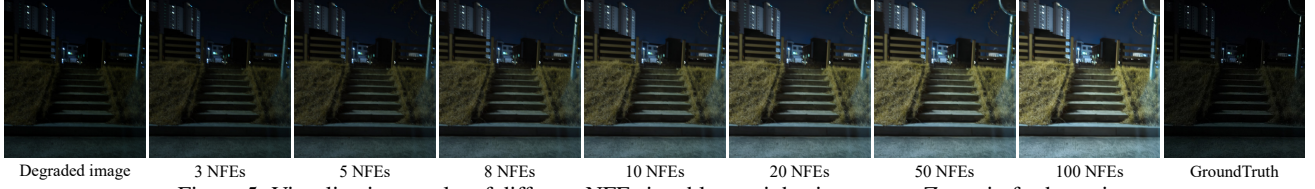


Figure 5. Visualization results of different NFEs in a blurry night-time scene. Zoom in for best view.

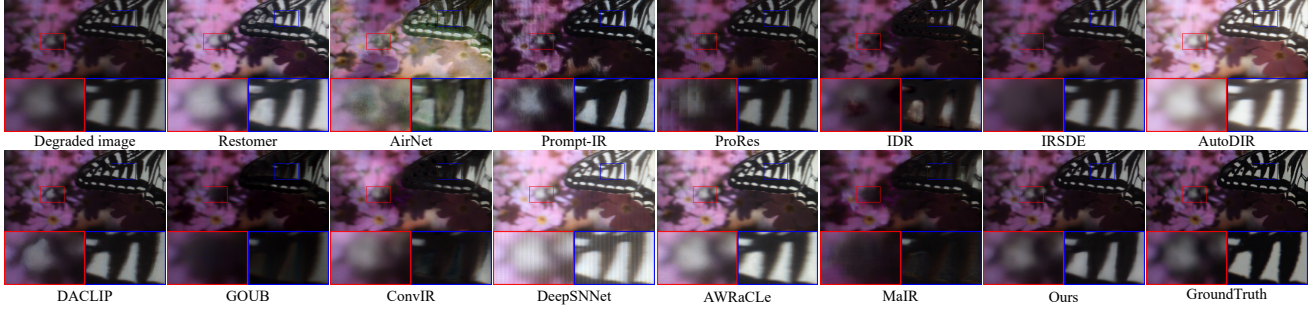


Figure 6. Visualization results of zero-shot generalization in real-world TOLED dataset. Zoom in for best view.

Table 5. Restoration performance of different sampling steps.

NFE	Deraining PSNR↑ SSIM↑	Enlightening PSNR↑ SSIM↑	Desnoising PSNR↑ SSIM↑	Dehazing PSNR↑ SSIM↑	Deblur PSNR↑ SSIM↑	Average PSNR↑ SSIM↑
2	26.06 0.790	14.82 0.648	20.28 0.835	17.20 0.809	28.06 0.855	22.81 0.815
5	30.05 0.876	23.35 0.809	30.47 0.947	29.01 0.929	29.13 0.878	29.61 0.905
10	30.31 0.884	24.53 0.812	32.60 0.961	33.45 0.965	29.04 0.877	31.04 0.917
20	30.10 0.882	24.35 0.811	31.96 0.959	32.25 0.961	28.94 0.875	30.58 0.915
50	29.92 0.880	24.21 0.809	31.59 0.958	31.56 0.958	28.85 0.873	30.28 0.913
100	29.84 0.879	24.13 0.808	31.49 0.957	31.39 0.957	28.80 0.873	30.19 0.912

Table 6. Restoration performance of different π .

π	Deraining PSNR↑ SSIM↑	Enlightening PSNR↑ SSIM↑	Desnoising PSNR↑ SSIM↑	Dehazing PSNR↑ SSIM↑	Deblur PSNR↑ SSIM↑	Average PSNR↑ SSIM↑
0	28.10 0.841	19.68 0.722	30.24 0.927	28.13 0.936	26.58 0.806	28.21 0.872
1	29.56 0.876	21.71 0.749	32.79 0.957	30.74 0.961	27.66 0.838	30.15 0.903
$x_0 - x_T$	30.31 0.884	24.53 0.812	32.59 0.961	33.45 0.965	29.04 0.877	31.04 0.917
$ x_0 - x_T $	30.36 0.883	24.37 0.812	32.40 0.957	33.19 0.965	28.99 0.876	30.94 0.915

ing accuracy gains from additional iterations. Beyond this threshold, performance gradually declines as NFEs rise. The underlying reasons are that our model is designed to handle diverse degradation types within a unified framework. In scenarios where samples exhibit multiple degradations, the model tends to prioritize the removal of the primary degradation before addressing secondary ones. Consequently, the restored output may deviate from the available reference, as shown in Fig 5. In conclusion, we adopt 10 sampling steps to ensure performance and efficiency.

Impact of diverse diffusion bridge settings. By appropriately selecting π , our method can establish equivalence with other diffusion bridges. Hence, we perform the restoration performance comparisons with different π selections, as presented in Tab. 6. The model is akin to flow matching as $\pi = 0$, yielding moderate results. It resembles stochastic interpolants and performs better as $\pi = 1$. Configuring π as the distributional residual or its absolute value is our formulation. These two variants produce similar results and achieve the best overall performance, thus verifying that residual bridge score matching offers a robust and effective paradigm for universal image restoration.

Table 7. Comparison under unknown tasks setting (under-display camera image restoration) on POLED and TOLED datasets.

Method	POLED				TOLED			
	PSNR↑	SSIM↑	MSE↓	LPIPS↓	PSNR↑	SSIM↑	MSE↓	LPIPS↓
Restormer [78]	11.500	0.445	0.077	0.494	11.094	0.495	0.106	0.330
AirNet [29]	5.705	0.103	0.324	1.072	9.706	0.530	0.117	0.403
Prompt-IR [51]	11.589	0.429	0.075	0.541	13.088	0.504	0.105	0.336
ProRes [44]	10.284	0.433	0.103	0.473	28.452	0.834	0.002	0.212
IDR [82]	13.583	0.466	0.057	0.551	24.259	0.759	0.008	0.253
IRSDE [41]	16.983	0.615	0.029	0.475	27.163	0.811	0.002	0.243
AutoDIR [21]	8.627	0.404	0.151	0.406	9.354	0.443	0.130	0.338
DA-CLIP [42]	16.788	0.559	0.025	0.469	27.256	0.789	0.003	0.201
GOUB [75]	12.922	0.525	0.053	0.446	23.177	0.761	0.007	0.269
ConvIR [11]	9.370	0.429	0.130	0.477	13.659	0.558	0.091	0.316
DeepSNNNet [12]	10.195	0.411	0.113	0.534	17.394	0.576	0.075	0.303
AWRaCLE [53]	11.208	0.431	0.091	0.513	10.540	0.495	0.121	0.331
MaIR [31]	11.072	0.423	0.086	0.529	23.637	0.770	0.005	0.271
RDBM	19.834	0.715	0.012	0.351	30.809	0.870	0.001	0.202

5.5. Zero-Shot Real-world Generation

To evaluate the generalization ability of our method, we do zero-shot generalization for unknown and known restoration tasks in real-world scenes. “Unknown” denotes cases where the degradation type is unspecified and may be compound, whereas “known” matches our task specification. As all methods are re-implemented on mixed datasets, they inherently handle diverse degradation types. In comparison, our method achieves strong performance in both settings.

Unknown task generalization. POLED and TOLED [86] are captured by under-display cameras in high-resolution with different degradation types, which fully meet the real-world scene. The quantitative results are reported in Tab. 7 while the visual comparisons are illustrated in Fig. 6. Evidently, our method achieves the best metric evaluation and our restored image is the most similar to ground-truth.

Known task generalization. As real-world datasets mainly have no ground truth, we use the non-reference metric, i.e., MetaIQA [87] and NIQE [48], to assess the perceptual qual-



Figure 7. Visualization results of image translation (top row) and image inpainting (bottom row). Zoom in for best view.

Table 8. Comparison under known task generalization setting.

Method	Deraising		Enhancement		Desnowing		Dehazing		Deblurring	
	MetalQA $\uparrow$	NIQE $\downarrow$	MetalQA $\uparrow$	NIQE $\downarrow$	MetalQA $\uparrow$	NIQE $\downarrow$	MetalQA $\uparrow$	NIQE $\downarrow$	MetalQA $\uparrow$	NIQE $\downarrow$
Restormer [78]	0.231	13.115	0.328	3.828	0.357	5.845	0.437	4.400	0.303	6.734
AirNet [29]	0.232	11.668	0.280	3.674	0.347	6.091	0.440	4.623	0.286	6.393
Prompt-IR [51]	0.232	11.439	0.308	3.797	0.361	5.840	0.437	4.962	0.286	6.670
ProRes [44]	0.226	13.110	0.348	3.933	0.355	5.976	0.434	5.444	0.297	6.574
IDR [32]	0.231	11.100	0.224	3.866	0.363	5.850	0.453	4.634	0.300	6.683
IRSDNet [41]	0.230	11.391	0.351	3.809	0.357	5.874	0.427	4.134	0.285	6.289
AutoDIR [21]	0.231	10.800	0.366	3.910	0.373	5.831	0.470	9.881	0.308	6.493
DA-CLIP [42]	0.232	10.604	0.334	3.720	0.361	5.864	0.460	6.531	0.310	6.058
GOUB [75]	0.231	11.566	0.373	3.928	0.360	5.853	0.458	4.104	0.278	6.303
ConvIR [11]	0.236	10.280	0.370	3.723	0.364	5.813	0.446	4.645	0.313	6.465
DeepSNNNet [12]	0.231	11.446	0.348	3.896	0.367	5.882	0.436	4.662	0.301	6.525
AWRACLE [53]	0.232	12.016	0.366	3.796	0.363	5.898	0.426	4.649	0.306	6.516
MaIR [31]	0.234	10.804	0.350	3.666	0.363	5.890	0.245	22.446	0.284	6.590
RDBM	0.238	9.559	0.397	3.663	0.396	5.482	0.483	3.973	0.343	5.671

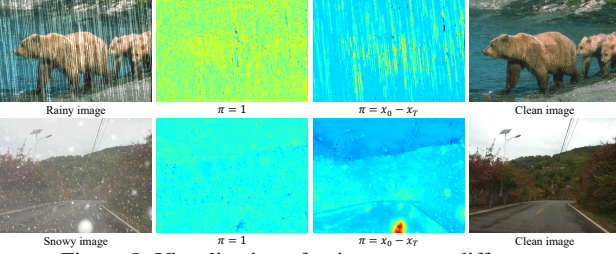


Figure 8. Visualization of noise maps on different π .

ity, as provided in Tab. 8. Results show that our method outperforms other universal models in various benchmarks and remains the most competitive.

5.6. Noise Maps Visualization

To further elucidate the superiority of our method, we visualize the predicted noise maps generated at a random time point in the reverse process of bridge models under different settings of π , as depicted in Fig. 8. Obviously, naive diffusion bridge ($\pi = 1$) blindly conducts global noise removal for the reconstruction of missing details. In contrast, our method ($\pi = x_0 - x_T$) performs adaptive restoration, as the noise maps are concentrated in degraded regions while remaining relatively smooth in non-degraded areas. In summary, our method can adaptively restore degradation in different regions, showcasing its high flexibility.

5.7. Image Translation and Inpainting

RDBM owns distinct advantages in mapping the data distribution to the prior distribution, thereby enabling extensive validation on similar computer vision tasks. To this end, we

Table 9. Quantitative results of image translation and inpainting.

Method	Image Translation [18]				Image Inpainting [22]			
	Edges $\rightarrow$ Handbags-256 $\times$ 256				Celebrate-HQ-256 $\times$ 256			
	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$
DDPM [63]	8.39	0.447	8.39	0.412	19.22	0.746	0.526	0.126
ReFlow [38]	10.46	0.442	5.76	0.374	23.53	0.822	0.307	0.149
BBDM [30]	14.75	0.635	7.46	0.248	20.36	0.653	0.386	0.169
I ² SB [35]	12.57	0.615	6.64	0.357	27.34	0.890	0.379	0.054
RDDM [37]	14.66	0.645	5.72	0.256	23.94	0.852	0.167	0.119
GOUB [75]	16.58	0.700	8.76	0.288	31.56	0.920	0.321	0.065
RDBM	19.26	0.738	5.38	0.224	37.88	0.965	0.147	0.031

expand our experimental settings on image-to-image translation and image inpainting to fully demonstrate the potential of our method. The former aims to transform an input image from one domain to another while preserving certain essential semantic or structural features. The latter focuses on filling in missing or corrupted regions within an image. Specifically, we adopt the widely used edge2handbags [18] dataset for image-to-image translation and apply the Celebrate-HQ dataset [22] with masks provided in [34] for image inpainting. All these datasets are scaled to 256×256 . We additionally employ Fréchet Inception Distance (FID) [15] for evaluation. Qualitative comparisons and quantitative results are presented in Fig. 7 and Tab. 9, respectively. Clearly, our method achieves the best visual effects and the best metrics assessments.

6. Conclusion

In this paper, we propose Residual Diffusion Bridge Model, termed as RDBM. Specifically, we theoretically reformulate the formulations of generalized diffusion bridge and derive the analytical formulas of its forward and reverse processes. Crucially, we leverage the residual from given distributions to dynamically modulate the probabilistic trajectories, thereby allowing the model to learn adaptive restoration of different regions with varying degradation levels. Furthermore, we unravel the fundamental mathematical essence of existing bridge models, and empirically verify the superiority of our models. Extensive experiments are conducted to demonstrate the state-of-the-art performance of our method across diverse tasks including image restoration, translation, and inpainting both qualitatively and quantitatively.

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